BEST BRAIN EXAMINATIONS KONSORTIUM
SPECIAL PRIVATE MOCK FOR BECE CANDIDATES – OCTOBER 2026
MARKING SCHEME – *SCIENCE*

PAPER TWO

SECTION A [40 Marks]

QUESTION ONE

(a) (i) **NUMBER OF PROTONS, NEUTRONS AND ELECTRONS IN THE ATOM**

Protons = 5
Neutrons= 6
Electrons=5
Electronic configuration = K2, L3

[2 marks @ ½ mark each]

(ii) **DIFFERENCES BETWEEN ELECTRONS AND PROTONS**

- Protons have positive electrical charges (+1) but electrons have negative electrical charges (−1).
- Protons reside in the nucleus of the atom, while electrons orbit the nucleus in electron shells.
- Protons determine the atomic number of an element and help identify the element but electrons are involved in chemical bonding and reactions, determining how atoms interact with each other.
- Protons are represented by the symbol p or p+ whiles electrons are represented by e−.
- Protons contribute to the atom's mass whereas electrons have minimal impact on atomic mass due to their small size.

[2 mark@ 1 mark each]

(iii) **ATOMIC NUMBER AND MASS NUMBER OF THE ATOM**

The atomic number is the number of protons in the atom which is equal to the number of electrons.
The atomic number is 5
Mass number is protons and neutrons in the atom.
Number of protons is =5, number of neutrons= 6
Therefore the mass number is 5+6 =11

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[2 mark@ 1 mark each]

(iv) **WHY THE ATOM CAN BE SAID TO BE ELECTRICALLY NEUTRAL.**

The equal number of positively charged protons in the nucleus (+5) and negatively charged electrons (-5) outside the nucleus makes the charges cancel out resulting in a net charge of zero.

[2 mark]

(v) **NAME AND SYMBOL OF THE ELEMENT WITH THE ATOMIC STRUCTURE ABOVE**

The atom is a Boron atom.
The symbol is B

[2 marks @ 1 mark each]

SUBTOTAL = 10 MARKS

(b) (i) **IDENTIFY THE TWO METHODS LABELLED X AND Y AND TYPES OF MAGNET**

X - Magnetization by Electrical Method
Y - Stroking method
X produces electromagnet
Y produces permanent magnet

[2 marks @ ½ mark each]

(ii) **DIFFERENCES BETWEEN THE TWO MAGNET TYPES PRODUCED.**

PERMANENT MAGNET	ELECTROMAGNET
permanently magnetised	temporarily magnetised
made of hard magnetic materials	made of soft magnetic materials
magnetism doesn't vary in strength	magnetism can be varied in strength according to needs
magnets' poles cannot be altered	magnets' poles can be modified

[2 marks @ 1 mark each]

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(iii) **STEPS INVOLVED IN USING THE METHOD LABELLED X**

- A coil of wire is wrapped around the metal (Eg, Iron nail
- A direct current (DC) is passed through the wire.
- The electric current creates a magnetic field that aligns the magnetic domains in the metal.

[2 marks]

(iv) **FACTORS THAT DETERMINE THE STRENGTH OF MAGNET PRODUCED AT X**

- The strength of the magnetic field
- The number of coils
-

[2 marks @ 1 mark each]

(v) **HOW THE METHOD Y IS PERFORMED**

Y - Stroking method

- This involves repeatedly rubbing a strong magnet along the metal in a single direction.
- This process aligns the magnetic domains in the metal, gradually magnetizing it.
- The more strokes, the stronger the resulting magnet.

[2 marks]

SUBTOTAL = 10 MARKS

(c) (i) **PRIMARY CONSUMERS**

- Rabbit
- Grasscutter
- Grasshopper

[2 marks @ 1 mark each]

(ii) **SIMPLE FOOD CHAINS**

- Grass → Rabbit → Hawk;
- Grass → grasshopper → Hawk
- Grass → grasscutter → snake → Hawk;
- Grass → grasshopper → lizard → Hawk
- Grass → grasscutter → Hawk

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[4 marks @ 2 marks each]

(iii) **FACTORS THAT CAN INTERRUPT THE BALANCE OF THE ECOSYSTEM**

- Deforestation
- Pollution
- Climate change

Overhunting

[2 marks @ ½ mark each]

(iv) **EFFECT OF THE ABSENCE OF I (PRODUCER) ON II (CONSUMER)**

The absence of producers (e.g., plants) would lead to a lack of food for primary consumers, causing a collapse in the food chain and ecosystem.

[2 marks]

SUBTOTAL = 10 MARKS

(d) (i) **DIAGRAMS OF THE SEEDBEDS**



(α) Raised seed bed;



(β) Sunken seed bed

[4 marks @ 2 marks each]

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(d) (i) **CALCULATING FORCE APPLIED**

Using Newton's Second Law (F=ma) [1 mark]

$F = 120\ kg \times 0.5\ m/s^2$ [1 mark]

$= 60\ N$ [1 mark]

(ii) **CALCULATING CHANGE IN FORCE**

New force (F') $= 120\ kg \times 0.3\ m/s^2$ [1 mark]

$F' = 36\ N$ [1 mark]

Change in force: $\Delta F = 60\ N - 36\ N = 24\ N$ [1 mark]

SUBTOTAL = 20 MARKS

QUESTION FIVE

(a) (i) **DEFINITION OF DECOMPOSITION**

Decomposition is the process by which organic matter is broken down into simpler compounds by decomposer organisms. [2 marks]

(ii) **ROLE OF DECOMPOSITION THE CARBON CYCLE**

It plays an important role in the carbon cycle by releasing carbon back into the atmosphere as carbon dioxide, which can be reused by plants through photosynthesis, and by storing some of the decomposed organic matter as organic carbon in the soil. [2 marks]

(b) (i) **HOW IGNEOUS AND SEDIMENTARY ROCKS ARE FORMED**

Igneous Rock: Involves the cooling and solidification of molten rock (magma or lava). When magma fro within the earth erupt onto the surface cools down, it crystallizes to form igneous rocks. [2 marks]

Sedimentary Rock: This process involves **compaction and cementation** (lithification) of sediments. Layers of sediments accumulate and the weight of the overlying sediments compact. [2 marks]

(ii) **DIFFERENCES BETWEEN IGNEOUS ROCK AND SEDIMENTARY ROCK.**

IGNEOUS ROCK	SEDIMENTARY ROCK
Forms from cooling and solidifying of molten magma	Forms from compaction and cementation of sediments
Typically has an crystalline texture.	Non-crystalline
Generally do not show layering.	Characterized by layers or strata
Do not contain fossils	Fossils are preserved, as organisms can be buried.

[2 marks @ 1 mark each]

(c) (i) **NEWTON'S SECOND LAW OF MOTION**

Newton's Second Law states that the acceleration of an object is directly proportional to the net force acting on it and inversely proportional to its mass. Force equals mass times acceleration $F = ma$, where F is the net force, m is the mass of the object, and a is its acceleration [2 marks]

(ii) **THE LAW OF MAGNETISM**

Like poles repel and opposite poles attract. [2 marks]

(iii) **THE LAW OF CONSERVATION OF ENERGY**

Energy cannot be created nor destroyed but can be converted from one form to another. [2 marks]

(d) (i) **STEPS TO SEPARATE A MIXTURE OF SUGAR AND CORN FLOUR**

1. Add water to the mixture to dissolve the sugar
2. Filter the mixture to separate the corn flour from the sugar solution
3. Boil the sugar solution to evaporate the water and obtain the sugar crystals

[3 marks @ 1 mark each]

(ii) **PROCESS INVOLVED IN EACH OF THE STEPS STATED ABOVE**

1. Solution/ dissolution
2. Filtration
3. Evaporation

[3 marks @ 1 mark each]

SUBTOTAL = 20 MARKS

TOTAL FOR SECTION B = 60 MARKS

THUS, TOTAL FOR PAPER 2 = (PART I + PART II) = 100 MARKS

END MARKING SCHEME

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PAPER I [40 MARKS]

1. B	11. D	21. B	31. B
2. D	12. C	22. D	32. A
3. B	13. A	23. D	33. B
4. B	14. A	24. C	34. D
5. A	15. C	25. B	35. C
6. A	16. A	26. B	36. B
7. C	17. D	27. A	37. B
8. A	18. C	28. B	38. D
9. B	19. B	29. B	39. B
10.B	20. B	30. A	40. D

GRAND TOTAL = PAPER 1 (40) + PAPER 2 (100) = 140 MARKS

$$\text{OVERALL SCORE} = \frac{\text{TOTAL SCORE}}{140} \times 100$$

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(ii) **REARRANGING IN SEQUENTIAL ORDER THE STEPS IN BED PREPARATION**

III- Clear the area of debris, weeds, and other unwanted plant material
II- Break up soil, remove weeds, and create furrows for planting.
IV- Level soil, remove rocks and debris, and create a smooth seedbed depth.
I- Water the soil thoroughly to ensure that it is evenly moist.

OR: III → II → IV → I

[2 marks @ ½ mark each]

(iii) **NAMING ONE TOOL USED AT EACH STEP IN BED PREPARATION**

III-Cutlass, Hoe, Garden Fork, Shovel
II- Hoe, Cultivators, Tiller, Garden Fork
IV- Rake, Shovel, Rollers, Harrows
I - Watering Can

[2 marks @ ½ mark each]

(iv) **BENEFITS OF PLANTING CROPS ON A SEED BED**

- Improved drainage
- Better soil structure
- Enhanced soil warming
- Reduced soil erosion
- Increased productivity

[2 marks @ 1 mark each]

SUBTOTAL = 10 MARKS

TOTAL FOR PART I = 40 MARKS

SECTION B [60 Marks]

QUESTION TWO

(a) (i) **SUGGESTED FARMING SYSTEM**

Mixed cropping - farming system where two or more crops are grown simultaneously on the same field.

[1 mark]

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(ii) **ADVANTAGES OF THE SUGGESTED FARMING SYSTEM**

- Reduces risk of total crop failure.
- Improves soil fertility.
- Promotes biodiversity.
- Utilizes resources efficiently.
- Suppresses weeds.
- Stabilizes yield.
- Diversifies income sources.
- Enhances crop quality.
- Maximizes land use.
- Distributes labor evenly.

[4 marks @ 1 mark each]

(b) (i) **NAMING OF STAGES AND FEATURES OF EACH IN THE LIFE CYCLE OF A MOSQUITO**

Adult

-has a slender body with two wings, six legs, and a long, slender proboscis that it uses to feed
- Adult mosquitoes mate and lay eggs, starting the life cycle over again.
- Adult mosquitoes can live for several weeks or months.

[1 mark for any one correct]

Pupa

- The pupa is also aquatic and lives in water.
- It has a comma-shaped body and does not feed.

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- At pupal stage, the mosquito undergoes metamorphosis, changing from larva to an adult.
- The pupal stage typically lasts a few days.

[1 mark for any one correct]

Eggs

- They are typically less than 1mm in size.
- They are usually oval-shaped and have a hard outer shell that protects the larva.

[1 mark for any one correct]

Larva

- Mosquito larvae are aquatic and live in water.
- They have a slender, worm-like body with a distinct head and thorax.
- They breathe through a tube called siphon, which helps get air at the surface of the water.
- They feed on organic matter in the water and molt several times as they grow.

[1 mark for any one correct]

(ii) **WAYS TO PREVENT THE BREEDING OF MOSQUITOES**

- Eliminating standing water where mosquitoes can lay their eggs
- Using insecticides to kill adult mosquitoes
- Using mosquito nets to prevent bites
- Wearing long-sleeved clothing to prevent mosquito bites

[2 marks @ 1 mark each]

(c) (i) **HOW A CAPACITOR DIFFERS FROM A BATTERY**

Capacitors and batteries are both energy storage devices, but they differ in how they store and release energy. Capacitors store energy in an electric field, while batteries store energy chemically.

[2 marks]

(ii) **APPLICATION OF CAPACITORS**

- Flash cameras
- Uninterruptible power supplies (UPS)
- Electric vehicles
- Power factor correction
- Audio equipment
- Defibrillators
- Laser systems
- Electric fences
- Hybrid energy systems
- Electronic toys

[3 marks @ 1 mark each]

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(d) (i) **DESCRIPTION OF A SUSPENSION**

A suspension is a mixture in which small solid particles are dispersed throughout a liquid and settle over time.

[2 marks]

(ii) **EXAMPLES OF SUSPENSIONS**

- Blood
- Paint
- Milk
- Fog
- Smoke
- Sand in water
- Flour in water
- Chalk in water
- Clay in water
- Muddy water
- River water after heavy rainfall

[2 marks @ ½ mark each]

SUBTOTAL = 20 MARKS

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QUESTION THREE

(a) (i) **EXAMPLES OF BIODEGRADABLE MATERIALS**

- Fruits
- Vegetables
- Flowers
- Plants
- Animals
- Paper
- etc.

[1 mark @ ½ mark each]

EXAMPLES OF NON-BIODEGRADABLE SUBSTANCES

- Plastics
- Chemicals
- Rrubber
- Paints
- Batteries
- Metals.

[1 mark @ ½ mark each]

(ii) **PRACTICES INVOLVED IN SUSTAINABLE WASTE MANAGEMENT**

- Reducing the amount of waste generated
- Recycling and recovering materials that can be processed and transformed into new products
- Composting – involves natural decomposition of organic waste, such as food scraps and yard waste
- Disposing waste off in a landfill
- Converting waste into energy, such as electricity or heat

[2 marks @ 1 mark each]

(iii) **WAYS PLANTS HELP MAINTAIN THE BALANCE OF CO_2 IN THE ATMOSPHERE**

- Plants absorb carbon dioxide during photosynthesis, converting it into glucose and oxygen.
- Plants act as carbon sinks, storing carbon in their biomass, reducing CO_2 levels.
- In photosynthesis, plants release oxygen, supporting atmospheric balance.
- Dead plants return some stored carbon to the atmosphere during decomposition.
- Plants provide habitats and food for soil microorganisms involved in carbon cycling

[2 marks @ 1 mark each]

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(b) (i) **DISTINCTION BETWEEN A SIMPLE MACHINE AND A COMPLEX MACHINE**

A simple machine is a device that makes work easier by changing the direction or magnitude of a force. A complex machine is a combination of two or more simple machines that work together to make work easier.

[2 marks]

(ii) **USES OF MACHINES**

(α) Escalators are moving staircases making easy movement between different floors feasible
(β) A crane machine is used for lifting heavy materials
(γ) Bicycles are widely used for transportation, recreation, and sport

[3 marks @ 1 mark each]

(c) (i) **TEETH FOUND IN MAMMALS WITH THEIR FUNCTIONS.**

Incisors: Front teeth, chisel-shaped, used for cutting and biting food.
Canines: Pointed, cone-shaped teeth, next to incisors, used for tearing and piercing food.
Premolars: Have flat surfaces with cusps, used for crushing and grinding food.
Molars: used for crushing and grinding food.

½ each for naming, 1 mark each for correct usage = **[6 marks @ 1½ mark each]**

(d) **USES OF GARDENING TOOLS**

(i) Hand fork – for loosening and turning soil, removing weeds, aerating the ground
(ii) Trowel – for digging small holes, transplanting seedlings, for planting small plants
(iii) Knapsack sprayer- For spraying pesticides, herbicides and weedicides, Spraying liquid fertilizers

[3 marks @ 1 mark each]

SUB-TOTAL = 20 MARKS

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QUESTION FOUR

(a) (i) **MEANING OF ANIMAL FEED**

An animal feed is any foodstuff or combination of foodstuffs, containing essential nutrients, given to domestic animals (livestock) to promote their growth, production and health.

OR: Animal feed refers to food material, processed or raw, given to domestic animals, designed to provide the necessary nutrients for growth, maintenance, production and reproduction.

[1 mark]

(ii) **LIST FOUR DIFFERENT VARIETIES OF FEED USED IN FARMING**

- Hay
- Silage
- Grains
- Protein Supplements
- Bran
- Molasses
- Pre-mixed feeds
- Mineral Blocks
- Vitamins
- Probiotics
- Pasture
- Crop Residues

[4 marks @ 1 mark each]

(b) (i) **STRUCTURES FOUND IN THE CYTOPLASM**

- Vacuole
- Mitochondrion
- Chloroplast
- Golgi apparatus
- Ribosomes
- Endoplasmic reticulum
- Lysosomes
- Plasmid

[2 marks @ 1 mark each]

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(ii) **FUNCTIONS OF THE CYTOPLASM OF A CELL**

- It provides a medium for the organelles to carry out their functions
- It also helps with cell shape and movement.
- It stores macromolecules
- It acts as the host to metabolic processes
- Its responsible for holding the components of the cell.

[3 marks @ 1 mark each]

(c) **DIFFERENCES BETWEEN SALT SOLUTION AND DISTILLED WATER**

- Salt solution is a mixture while distilled water is a compound
- The salt and water composition of the salt solution are not in any fixed ratio while the chemical composition of distilled water is fixed
- The salt and water components of the salt solution is easily separated by physical means but the composition of the distilled water cannot be separated by physical means
- The properties of distilled water differ from that of the constituent elements while both water and salt retain their properties in the salt solution
- Distilled water is a pure substance while salt solution is an impure substance.
- Salt solution can be formed by a simple physical combination of salt and water while distilled water can only be formed by chemical combination of hydrogen and water.

[4 marks @ 2 marks each]

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